WORKFORCE OS

Enterprise Workforce Management, LMS, Outsourcing & AI Engine

Comprehensive End-to-End Quality Assurance, Security Audit & Handoff Testing Manual

Release: v1.0.0-PROD | Target Branch: dev | Architecture: DEDICATED Tenancy | Date: November 2026

1. Executive Testing Scope & Architecture

This document serves as the formal Quality Assurance (QA) and Engineering Handoff Manual for the WorkforceOS platform module in Ofia ERP. It covers complete testing flows across all 10 completed implementation phases, encompassing 14 administration consoles, 4 public/client token-gated portals, the underlying PostgreSQL dynamic tenancy engine (getTenantSql), and AI operational services. All test cases verify end-to-end functionality, strict database isolation, and state persistence.

User Role / Persona	Primary Experience & Console	Permissions & Functional Scope
Super Admin / Org Admin	/erp/admin/workforce	Full platform governance, departments, branches, audit logs, and global KPIs.
People Chief / Recruiter	/erp/admin/workforce/recruitment	Job openings, candidate pipeline, scorecards, interviews, and digital offers.
Academy LMS Instructor	/erp/admin/workforce/academy	Curriculum authoring, modules, assessments, exam grading, and certifications.
Line Manager / Supervisor	/erp/admin/workforce/tasks	Project management, task assignments, daily work log approval, and performance PIPs.
Operations Lead	/erp/admin/workforce/deployment	Capacity planning, 12-month Gantt availability, roll-offs, and staffing requisitions.
Embedded Staff / Intern	/erp/employee	Clock-in/out, work logs, LMS coursework, practical tasks, and personal dossier.
Corporate Client Lead	/erp/client	Restricted outsourcing portal: assigned specialists, SLA tickets, timesheet sign-off & CSAT.

2. Comprehensive Phase-by-Phase Test Suites

Phase 1: Organizational Hierarchy & Departments	Route: /erp/admin/workforce/departments
Test TS-01-DEPT: Department CRUD & Cost Center Budget	1. Navigate to /erp/admin/workforce/departments. 2. Click '+ Add Department', input 'Infrastructure & Cloud', Code 'INFRA', Head 'Dr. Tunde Fashola', Budget 15,000,000 NGN. 3. Verify department card appears with badge, staff count, and persistent DB record. 4. Click 'Edit', update budget to 18,500,000 NGN. Refresh page; confirm budget persists. Pass Criteria: Record created in workforce_departments table with tenant_slug isolation; audit log logged.
Test TS-01-BRANCH: Branch Creation & Headquarters Flag	1. Select 'Branches' tab; click '+ Add Operational Branch'. 2. Input Name 'Abuja Regional Office', City 'Abuja (FCT)', Address 'Plot 12 Central Business District', Staff Count 12. 3. Set 'Headquarters' flag to false. Click Save. Pass Criteria: Record persisted in workforce_branches; correctly differentiated from Lagos HQ.
Test TS-01-ORG-TREE: Visual Dynamic Org Hierarchy Tree	1. Select 'Organization Hierarchy Tree' tab. 2. Verify root node 'Executive Management & Directorate' renders with subordinates. 3. Expand 'Technology Directorate' -> 'Platform Engineering' -> 'Security & QA'. Pass Criteria: Tree expands dynamically without layout overflow; department heads accurately labeled.

Phase 2: 14-Tab Personnel Dossier & Document Vault	Route: /erp/admin/workforce/people/[id]
Test TS-02-DOSSIER: 14-Tab Dossier Navigation	1. Navigate to /erp/admin/workforce/people and select personnel 'Kalu Uche' (per-01). 2. Click through all 14 tabs: Profile, Employment, Org & Reporting, Compensation, Deployment, LMS Academy, Tasks, Attendance, Appraisals, Documents, Assets, Compliance, Emergency, Audit Trail.

Phase 2: 14-Tab Personnel Dossier & Document Vault	Route: /erp/admin/workforce/people/[id]
	Pass Criteria: Each tab renders dedicated data models; no 404 or state desync; smooth transitions.
Test TS-02-DOC-VAULT: Cryptographic Document Upload & Verification	1. In 'Documents' tab, click '+ Upload Verified Document'. 2. Select type 'National Identity Number (NIN)', Doc Number 'NIN-992817291', File URL. 3. Verify document appears with status 'pending_verification'. Click 'Verify Document'. Pass Criteria: Status mutates to 'verified' with green badge, timestamp, and audit log recorded in DB.

Phase 3: Recruitment, Interview Room & Digital Offer Portal	Route: /erp/admin/workforce/recruitment
Test TS-03-KANBAN: Candidate Pipeline Stage Drag & Transition	1. Navigate to /erp/admin/workforce/recruitment. 2. Drag candidate 'Femi Adebayo' from 'Applied' to 'Shortlisted', then to 'Technical Interview'. 3. Inspect database: verify workforce_candidates.stage is updated to 'technical_interview'. Pass Criteria: Stage updates reactively and persists across browser refreshes.
Test TS-03-INTV-ROOM: Public Tokenized Interview Room	1. Click 'Schedule Interview' for candidate. Choose Date, Google Meet URL, Panel Members. 2. Copy generated tokenized URL: /erp/admin/workforce/recruitment/interview/[token]. 3. Open URL in an incognito window without ERP authentication. Pass Criteria: Interview room renders candidate name, role, countdown timer, agenda, and meeting button.
Test TS-03-OFFER-PORTAL: Digital Offer Issuance & Signature	1. In candidate modal, click 'Issue Offer'. Set designation 'Associate Engineer', salary 280,000 NGN. 2. Copy offer tokenized link: /erp/admin/workforce/recruitment/offer/[token]. Open in incognito. 3. Candidate reviews formal letter, types digital signature 'Femi Adebayo', and clicks 'Accept Offer'. Pass Criteria: Offer status transitions to 'accepted'; candidate stage mutates to 'offer_accepted'; audit logged.

Phase 4: Automated Onboarding & User Provisioning	Route: /erp/admin/workforce/onboarding
Test TS-04-DOSSIER-PORTAL: Candidate Intake Self-Service	1. From recruitment or onboarding console, generate Onboarding Dossier link for accepted candidate. 2. Open /erp/admin/workforce/onboarding/portal/[token] in incognito window. 3. Input BVN, NIN, Residential Address, Bank Name, Account Number, Next of Kin details. 4. Submit dossier. Verify success screen confirms submission to People Operations. Pass Criteria: Packet status updates to 'submitted' with all encrypted PII stored in workforce_onboarding_packets.
Test TS-04-PROVISION: 1-Click System User & Staff Account Activation	1. Log into /erp/admin/workforce/onboarding as HR Administrator. 2. Inspect submitted packet. Verify all required KYC documents are attached. Click 'Approve Dossier'. 3. Click 'Provision System Account'. Confirm modal with corporate email generation. Pass Criteria: System generates per-[id]; inserts active staff into workforce_people; sets candidate stage to 'onboarded'; updates packet to 'provisioned'.

Phase 5: Talent Academy LMS & Cryptographic Certifications	Route: /erp/admin/workforce/academy
Test TS-05-SYLLABUS: Modular Course Curriculum & Lessons	1. Navigate to /erp/admin/workforce/academy; select course 'Enterprise TypeScript & Microservices'. 2. Click '+ Add Module'. Add Module 4: 'Event-Driven Kafka Architecture & WebSockets'. 3. Add 2 lessons with duration and markdown content. Save. Pass Criteria: Modules and lessons persist in workforce_course_modules with strict order indexing.
Test TS-05-CERT: Cryptographic Certificate Verification	1. Issue certificate to completed learner. Note generated SHA-256 verification hash. 2. Navigate to public verification route: /erp/admin/workforce/academy/verify/[hash]. Pass Criteria: Cryptographic verification portal displays verified badge, candidate name, course title, issue date, and security seal.

Phase 6: Project Entities, Task Dependencies & Deliverable Vault	Route: /erp/admin/workforce/tasks
Test TS-06-PROJECT: Project Creation & Budget Burn Tracking	1. Navigate to /erp/admin/workforce/tasks, select 'Projects' tab; click '+ Create Project'. 2. Input Title 'Mainland Courier WebSocket Gateway', Client 'Prime Logistics Ltd', Budget 6,000,000 NGN. 3. Assign Lead 'Kalu Uche'. Click Create Project. Pass Criteria: Project persists in workforce_projects; linked to client; tasks and deliverables count initialized.
Test TS-06-DELIVERABLE: Artifact Upload & Supervisor Sign-off	1. On task detail, click 'Submit Deliverable'. Upload OpenAPI schema v1.2 specification. 2. Log in as Line Supervisor. Open deliverable review modal; approve with remarks 'Approved for deployment'. Pass Criteria: Status mutates to 'approved'; reviewed_by and reviewed_at persisted in workforce_deliverables.

Phase 7: Performance Appraisals, 360° Scorecards & PIPs	Route: /erp/admin/workforce/performance
Test TS-07-360-SCORECARD: Multi-Rater Scorecard Submission	1. Navigate to /erp/admin/workforce/performance. Select active cycle 'Q2 2027 Appraisal Cycle'. 2. Click '+ Submit Scorecard'. Select Person 'Chinedu Eze', Reviewer Type 'supervisor'. 3. Set Technical Competence: 5/5, Leadership: 4/5, SLA Adherence: 5/5, Culture: 5/5. 4. Submit scorecard. Confirm overall score calculates to 95%. Pass Criteria: Scorecard recorded in workforce_scorecards; cycle completed reviews count increments.
Test TS-07-PROMOTION: Automated Promotion & Salary Elevation	1. Navigate to 'Promotions & Progression' tab. Click '+ Nominate Promotion'. 2. Nominate 'Chinedu Eze' from 'Associate Systems Specialist' to 'Senior Systems Specialist'. 3. Approve nomination as Directorate Lead. Pass Criteria: Mutates workforce_people.designation; updates compensation tier; records PROMOTION_APPROVED audit log.

Phase 8: Workforce Planning, 12-Month Gantt & Capacity Matrix	Route: /erp/admin/workforce/deployment
Test TS-08-GANTT: 12-Month Interactive Availability Matrix	1. Navigate to /erp/admin/workforce/deployment/calendar. 2. Verify staff members (Kalu Uche, Chinedu Eze, etc.) display horizontal monthly bar allocations across Q1-Q4. 3. Click allocation bar for 'Prime Logistics'. Drawer opens displaying contract dates, allocation %, and monthly billing. Pass Criteria: Visual Gantt accurately reflects active date spans; drawer roll-off button active.
Test TS-08-APPROVAL-GATE: Multi-Stage Deployment Authorization	1. Navigate to /erp/admin/workforce/deployment, tab 'Directorate Approval Gate'. 2. Submit new deployment requisition in 'pending_approval' status. 3. Click 'Authorize Deployment'. Input approver name 'Directorate Lead'. Pass Criteria: Status mutates to 'approved'/'active'; staff status shifts from 'bench' to 'deployed'.

Phase 9: Restricted Client Portal for Outsourcing	Route: /erp/client
Test TS-09-ISOLATION: Corporate Client Context Switcher	1. Open /erp/client. Switch corporate dropdown between 'Prime Logistics Ltd' and 'Apex Academy Lagos'. 2. Verify roster of specialists, tickets, timesheets, and CSAT strictly filter to the selected client only. Pass Criteria: Zero data leakage between corporate clients; strict multi-tenant client boundary.
Test TS-09-TICKET: Client SLA Incident Desk with Countdown Clocks	1. In 'SLA Service Desk' tab, click '+ Log Incident Ticket'. 2. Enter subject 'API Timeout on Courier WebSocket Cluster', Priority 'P1' (Critical - 1h SLA). 3. Submit ticket. Verify table displays ticket with '60 mins' remaining countdown and P1 badge. Pass Criteria: Ticket inserted into workforce_tickets with assigned specialist; audit log logged.
Test TS-09-TIMESHEET: Corporate Timesheet Review & Sign-off	1. In 'Timesheets & Sign-Off' tab, locate daily work log submitted by Kalu Uche. 2. Verify logged hours (8.5 hrs) and summary. Click 'Sign Off Timesheet'. Pass Criteria: Button switches to 'Approved & Signed' green badge with stakeholder name and date; DB updated.
Test TS-09-CSAT: Monthly Quality CSAT Review Submission	1. In 'Quality CSAT' tab, click '+ Submit Monthly CSAT Review'. 2. Adjust sliders: Technical 5/5, Punctuality 5/5, SLA Adherence 5/5. Note computed 100% Quality Score. 3. Enter feedback 'Flawless execution'. Submit review. Pass Criteria: Review saved in workforce_csat_reviews; appears in quality review history.

Phase 10: AI Operations, CRM Sync & Multi-Channel Webhooks	Route: /erp/admin/workforce/ai-operations
Test TS-10-MATCHER: Natural Language Semantic Staff Matcher	1. Navigate to /erp/admin/workforce/ai-operations, tab 'Natural Language Staff Matcher'. 2. Click prompt chip: '3 available network engineers in Lagos with CCNA'. Click 'Run AI Match'. 3. Verify AI ranks bench staff by match percentage, highlighting matched skills, missing skills, and status. Pass Criteria: Semantic scoring executes without errors; top candidates accurately ranked.
Test TS-10-PARSER: AI CV / Resume Entity Extractor	1. Select 'AI Resume / Dossier Parser' tab. Click 'DevOps Sample' prefill button. 2. Click 'Parse Candidate Dossier'. Confirm extracted name, email, phone, experience, and fit score. 3. Click 'Import into Recruitment Pipeline'. Verify candidate appears in Recruitment Kanban. Pass Criteria: JSON successfully extracted; candidate inserted into workforce_candidates with score.
Test TS-10-BRIEFING: Executive Morning Briefing Synthesizer	1. Select 'Executive Morning Briefing' tab. Click 'Generate Live Briefing'. 2. Verify operational health score (e.g. 98%), executive summary, key highlights, and active SLA risks. Pass Criteria: AI synthesizes live work logs, tickets, and attendance into executive briefing payload.
Test TS-10-CRM-SYNC: CRM Won Deal Auto-Conversion to Requisition	1. Select 'CRM Sync & Notification Webhooks' tab. 2. Fill CRM deal: Company 'Zenith Microfinance Bank', Value '3,800,000 NGN', Owner 'Tunde Bakare'. 3. Click 'Simulate CRM Won Deal'. Pass Criteria: Client account upserted in workforce_clients; staffing requisition created in workforce_requests; webhook notification logged.

3. Master End-to-End Integration Flows

Flow #	Master Integration Lifecycle	Stages & Verification Milestones
MF-01	Full Talent Lifecycle: From Application to Certified Staff	1. Candidate submits application via recruitment intake. 2. Recruiter schedules interview -> public tokenized interview room verified. 3. Recruiter issues formal digital offer -> candidate signs electronically. 4. Candidate completes self-service onboarding dossier (BVN, NIN, Bank). 5. HR approves dossier -> 1-click system user provisioning executes. 6. Staff created in workforce_people with active ERP status. 7. Staff enrolls in Talent Academy -> completes syllabus -> takes exam. 8. Cryptographic certificate issued with verifiable SHA-256 hash.
MF-02	Commercial Outsourcing Contract Lifecycle: From Deal Won to Timesheet Sign-off	1. CRM deal closed-won -> auto-converts to client account & workforce request. 2. Operations manager assigns specialists on 12-Month Gantt Calendar. 3. Directorate Lead approves allocation via formal deployment gate. 4. Specialist status shifts from 'bench' to 'deployed'. 5. Specialist logs daily work hours on mainland delivery project. 6. Corporate client logs in at /erp/client -> reviews timesheet -> signs off. 7. Client logs P1 support incident -> specialist resolves within SLA. 8. Client submits monthly CSAT review (100% Quality Index).

4. Security, Compliance & QA Sign-Off Checklist

Checklist Item	Standard / Constraint	QA Verification Method	Status
Multi-Tenant Isolation	Strict DEDICATED DB per tenant; dynamic resolver getTenantSql.	Verify all DML queries explicitly append WHERE LOWER(tenant_slug) = ${cleanSlug}.	PASSED [VERIFIED]
Zero Unsplash Compliance	Absolute ban on Unsplash imagery across all portals and code.	Codebase scan git diff | grep -i 'unsplash' yields zero occurrences.	PASSED [VERIFIED]
Zero Hardcoded Credentials	Zero plaintext passwords, cluster URLs or secrets in code.	Audit of all route handlers and store modules confirms env/DB config.	PASSED [VERIFIED]
TypeScript Compilation	Strict type safety across types, store, API routes, and UI.	Full project compiler check npx tsc --noEmit yields 0 errors.	PASSED [VERIFIED]
Git Branching Governance	Target strictly branch dev; never merge directly to main.	All 10 phases committed in atomic units and pushed to origin/dev.	PASSED [VERIFIED]